

# Apiwit Karnjanavivin

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## Research Interests

Vertical co-design between high-level AI applications and low-level software infrastructure (operating systems and distributed systems) to optimize computational resource utilization and minimize execution time for production-grade RAG and agentic AI systems.

## Education

**Master of Science in Data Science, Artificial Intelligence, and Digital Business (M.Sc.) - First Class Honor** June 2024 – September 2025  
Gisma University of Applied Science  
Potsdam, Germany  
**Final Dissertation:** Better than it is: enhancing DeepSeekMoE architecture-based chatbots using dynamic routing algorithms

**Bachelor of Engineering in Industrial Engineering (B.Eng.)** August 2016 – April 2021  
Kasetsart University  
Bangkok, Thailand  
**Final Dissertation:** Designing of lean-based hydroponic farm, its business plan, and financial analysis

## Research Experience

**AI Research Assistant (Volunteer)** July 2025 – Present  
Gisma University of Applied Science  
Potsdam, Germany

- Proposed enhancing the DeepSeekMoE architecture by redesigning its routing algorithm, auxiliary load balance loss, and model training process to optimizing computational resource usage while retaining performance in training, fine-tuning, and inference.
- Co-authored a research paper targeted ACM publication, contributing introduction, foundations, related works, approach, and evaluation.
- Designed and executed ablation studies on the related routing algorithm, auxiliary load balance loss, and model architecture parameters on the prototype model with measurement metrics following research objectives to verify research claims and understand interactions between them.
- Contributing publication repository with ensure reproducibility for research paper publication of the baseline and prototype models with monitoring experiments, according to the objectives of the research.
- Collaborating on the research paper with a supervising professor through regular progress updates and iterative improvements based on professor’s feedbacks.

## Professional Experience

**In-Process Warehouse Leader** October 2021 – November 2023  
Raku Innovation Co., Ltd.  
Bangkok, Thailand

- Created weekly Material Requirements Planning (MRP) to align inventory with production targets and reported status to supervisors.
- Led a small team in daily warehouse operations, achieving team KPIs through effective tasks assignment and team working process optimization.
- Conducted regular audits to maintain ERP system accuracy for raw materials and work-in-process.
- Managed supply chain coordination with other internal teams and outsourcing partners to meet delivery deadlines.

## Academic Projects

- Analysis of AI and Algorithmic Recommendation Systems on Large Social Network Platforms Mini Research Paper
- K-Means Clustering to Image Quantization
- Deep Q-Learning Implementation and Hyperparameters Analysis
- LLMs Text Generation Approaches
- Adversarial Text Generation Against LLM Judges
- A Survey of LLMs Architectures
- A Survey of LLMs Fine-Tuning Optimization Approaches
- A Survey of Retrieval-Augmented Generation (RAGs)
- Hospital Stay Prediction - Machine Learning Operations (MLOps)
- Fine Tuning LLM - Large Language Model Operations (LLMOps)
- Retrieval Augmented Generation (RAG) - Large Language Model Operations (LLMOps)

Project documentation and code available at : <https://github.com/MisterFOURXXX/data-science-and-ai-academic-projects>

## ● Achievements

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- Social Media Extremism Detection Challenge - Top 24 percent of participants
- CIBMTR - Equity in post-HCT Survival Predictions Competition - Top 33 percent of participants
- PHEMS Hackathon: Pediatric Sepsis Prediction - Top 33 percent of participants
- Beyond Visible Spectrum: AI for Agriculture 2025 - Top 40 percent of participants
- FlightRank 2025: Aeroclub RecSys Cup - Top 45 percent of participants

Kaggle Competition

**Project documentation and code available at :** <https://github.com/MisterFOURXXX/kaggle-competitions>

## ● Training and Certifications

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### Programming and Data Science Fundamentals

Coursera

- Python 3 Programming – University of Michigan
- Advanced Statistical Techniques for Data Science – Illinois Tech
- Git and GitHub Complete Master Class – Packt
- Applied Python Data Engineering – Duke University
- IBM Data Engineer – IBM

### Machine Learning and Neural Network

- IBM Machine Learning – IBM
- IBM AI Engineering – IBM
- Foundations of Neural Networks – Johns Hopkins University
- Applied Machine Learning – Johns Hopkins University
- AI and Machine Learning Essentials with Python – University of Pennsylvania
- Machine Learning: Algorithms in the Real World – Alberta Machine Intelligence Institute

### Deep Learning

- TensorFlow 2 for Deep Learning – Imperial College London
- IBM Deep Learning with PyTorch, Keras and Tensorflow – IBM
- Data Analytics and Deep Learning – Illinois Tech

### Generative AI and Large Language Models (LLMs)

- Generative AI for Data Engineers – IBM
- Generative AI for Data Scientists – IBM
- Generative AI Engineering with LLMs – IBM
- IBM Generative AI Engineering – IBM
- IBM RAG and Agentic AI – IBM

### Operations (MLOps and LLMOps)

- MLOps | Machine Learning Operations – Duke University
- Large Language Model Operations (LLMOps) – Duke University

## ● Skills and Proficiencies

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### Programming Languages

- Python, Bash, SQL, Git, LaTeX (Overleaf)

### Data Science and Visualization

- Polars, Pandas, NumPy, Plotly, Seaborn, Matplotlib, Scikit-learn

### Machine Learning, Deep Learning, and Large Language Models

- Scikit-learn, PyTorch, TensorFlow/ Keras, Hugging Face, Unsloth, LangChain, LangGraph, vLLM

### Web Development

- HTML, CSS, JavaScript

### Systems, Infrastructure, and Operations

- MLflow, DagsHub, DVC (Data Versioning), Apache Airflow, AWS, Terraform, Docker, FastAPI, Streamlit

### Languages

- English (Fluent), Thai (Native), German (Beginner)

## ● References

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Gisma University of Applied Sciences, Professor

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Gisma University of Applied Sciences, Professor